# When Proofs Are Abundant, What Is a Paper For?

## Intension and Scientific Chains of Thought in the Age of AI Mathematics

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# Content

What is a scientific communication?

Clear Definitions

- Derivation-Level Intension
- Directional Extension

Example on 4 microstructure papers

- Domains to Directional Extension tells the path from equilibrium to local optima
- Derived-Level Intension shows how papers communicate





# Why asking the question today

- Big AI-Tech companies are attacking celebrated maths problems,

☞ a lot of buzz in the media, **potentially bad reputation for maths**.

- It is not LLM alone: they use a Lean harness, [Bansal 2019; Han 2021; survey: Jiang 2026].
  After the failure of LLM hallucinations, the fix of RAG was not enough, the agentic fix emerged, now the harness (towards more typing).

☞ **it may give the impression of a limitless progress**, but it is not a continuous advance of the same approach, it is a sequence of fixes.

- Nevertheless these tools are important and useful, lifting us towards an higher level of abstraction

☞ most of our work (and the one of our students) will be **hybrid** (like it was before? Google Scholar and Maple/Mathematica).

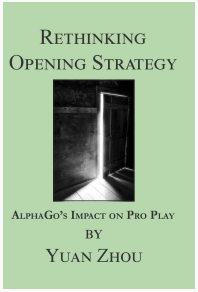
- Big AI-Tech companies push us to take a position with respect to 100%-made proofs,

☞ In my opinion **it is a trap**: we should focus on a way to communicate the results of hybrid work to continue to collectively to answer more problems a scientific way.





# Will LLMs Change What Counts As An Argument?




- Just as **AlphaZero** made Go players reconsider openings once thought to be losing [Zhou 2018], LLMs may reshape which steps of a chain of thoughts a community accepts as **familiar**, or **obvious**.
- In *The Sciences of the Artificial* [Simon 1969], Simon separates **understanding** from **acting**: a paper is a chain of reasoning *in action*.
- Anne Alombert [Alombert 2026]:

  ❝ *after the automation of practical skills,*
  *we are witnessing the automation of ~~thinking~~ reasoning* ❞

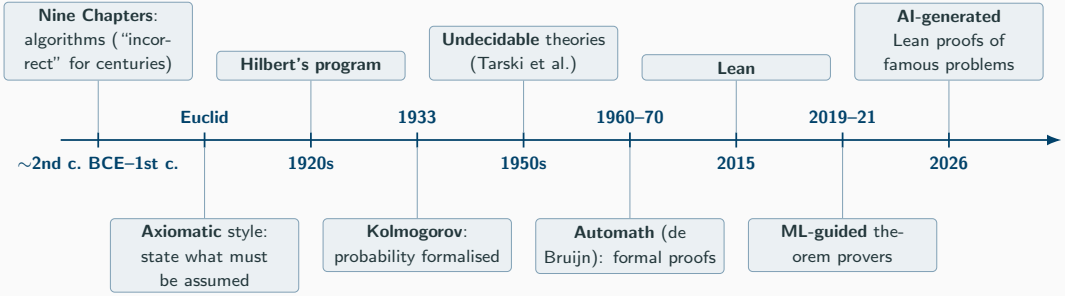
**Applied to a research paper:** the **act of reasoning** can be automated; the **act of understanding** is what a paper, and a talk, must keep conveying.

Nevertheless, the triptych:

**intension + scientific chain-of-thought + details of crucial components/atoms**

is not new.

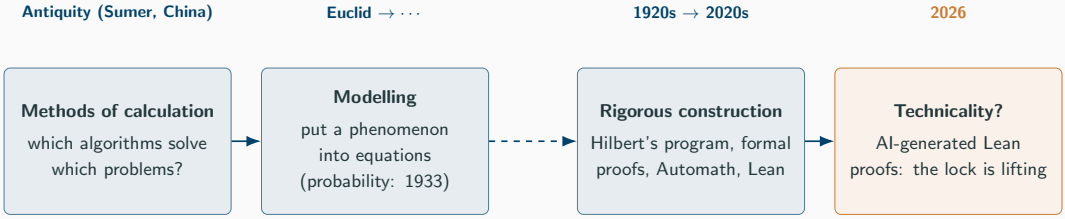
# The acceptance of a proof has always evolved




**China** The *Nine Chapters* were judged "incorrect mathematics" for centuries; since 1960–80 they are read as advanced *algorithms*, i.e. **constructive proofs** [Chemla–Shuchun 2004].

**Probability** Its formalisation was debated until **1933**. Undecidable statements then led to constructive and formal approaches [Zach 2020; de Moura 2015].

# A community works on what blocks its field, and this changes




Hilbert pointed out a century ago that

**rigorous constructions** were our bottleneck, but this is *recent*.

**When the bottleneck moves, the focus of the community moves with it.**

# What Remains Human, That Is Useful

- Research is **juggling** between the feasibility of components of a chain of thoughts and the design of this very chain, under the premises of an intension.
- How could a system *generate* an intension without a prior human-triggered one? Even a system that scans an ecosystem for problems **inherits the intension of those who built it**.
- Availability of AI **does not change this process**.

**A paper to be published should emphasise**

1. the **intension**,
2. the **scientific chain of thoughts**,

and make available **all elements** enabling to check carefully the components of the chain.





# Five Usages Of AI In A Research Paper: Risks & Reactions

intension $\rightarrow$ scientific chain-of-toughts $\rightarrow$ details on a handful of particular components

| Usage | What it does | The Risk & A Positive Reaction |
|---|---|---|
| Bilbio. search & summarise | Shortcuts to existing papers (Google Scholar vs. shelves). | **Risk:** Biases. Visibility strategies will adapt. |
| Prove components | Gets the proof of some steps of a given chain (short Lean scripts). | **Risk:** A chain of thought may become a black box if only the final proof is shared.<br>**Reaction:** **sharing the articulation of the scientific chain of thought** in papers is the better. |
| Explore chains of thoughts | Proposes plausible chains for a given intension. | **Risk:** The fixed cost for humans to build a reasonable chain vanishes, so **intensions must be judged very carefully**.<br>**Reaction:** The community must increase its effort to scrutinize the triptych: intension–reasoning chain–proofs. |
| Judge a reasoning | An LLM reviewer (e.g., your step 4); replaces working groups, co-authors, peers. | **Risk:** **Uniformisation** of production (as journals' editing already standardises).<br>**Reaction:** Journals can embrace this role explicitly. |
| Full production | Full delegation from intension to details. | 👍 **Never happened so far**. The intension will always belong to the humans: disclose the initial intension. |





# Next Section – What is a scientific communication?

What is a scientific communication?

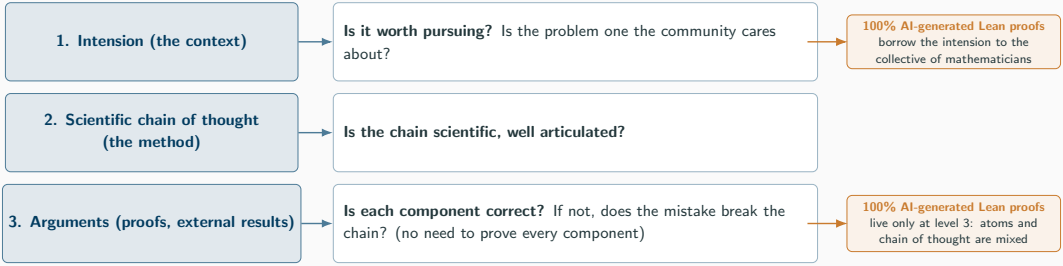
Clear Definitions

Example on 4 microstructure papers





# A paper or talk exposes the research to criticism at three levels




**Two usual criticisms of 100% Lean proofs** [Weatherall–Wolfson 2026]

- Lean scripts **convey no intension** (item 1).
- They **mix the chain of thought and the arguments** in atoms so small that no human can tell what is what (items 2 and 3).
  *"AI solved one of math's hardest problems. Humanity learned nothing (so far)"*

## A paper is a narrative exposed to criticism

Paper = intension + chain of thought + supporting arguments

- **Intension**: why this problem, from which angle
- **Chain of thought**: the reasoning steps, exposing weak and strong links
- **Arguments**: can be delegated to references, computations, Lean

Most published mathematics is discussed with referees on the first two, rarely on correctness alone. **Journals are not proof checkers**.

**Question**

In a world of abondance of proofs (i.e. if correctness becomes cheap), what is left for a paper to communicate?

Journals are overwhelmed by automated submissions; some suspended new evaluations.



# The message is simple: a scientific communication has two parts

## 1. The context

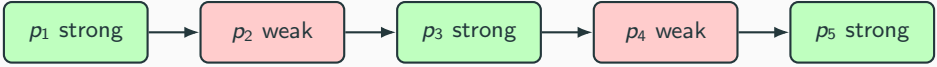
**formally: Derivation-Level Intension**

- Is it in the **target** of the community?
- Does it mobilise tools the community **feels able to judge**?

## 2. The method

**formally: Directional Extension**

- A chain cut into **segments of reasoning**
- **weak** segments limit the power of the result
- **strong** segments: to do better, change the approach; they can be **transported** to other contexts




the arguments behind each segment can be delegated: references, computations, Lean

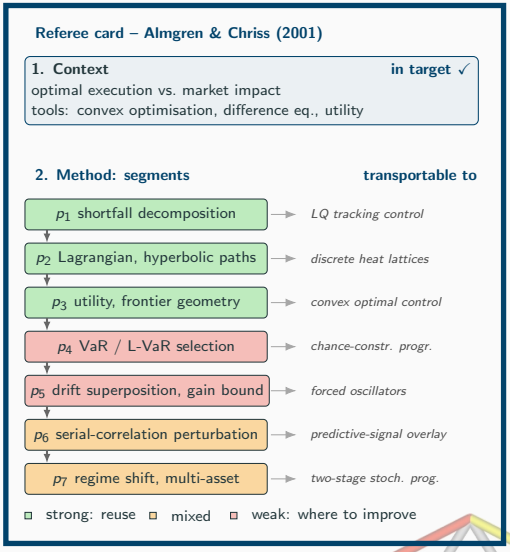
# What it gives to journals and to teaching

**Journals**, even with abundant proofs

- check the **fit of the Derivation-Level Intension**
- require an explicit **chain with weak and strong links, the Directional Intension**
- let machines check the arguments (Lean, references)

**Teaching**

- the shift from "a correct proof" to Derivation-Level Intension + Directional Intension usually happens *during the PhD*:
  $\Rightarrow$ **teach it earlier**




Example from the Almgren-Chriss Optimal Trading paper

## Why category theory, without letting it hide the message

The category-theoretic language carries two more messages:

1. **Rigour**: the two notions can be formulated precisely
   $\Rightarrow$ in principle they can be *extracted from a Lean script* (a research direction)

2. **Comparison**: we can become Grothendieck, *assisted by computers*
   $\Rightarrow$ comparative studies of proofs through their DLI/DE decompositions

**Illustration at the end of the talk**

Four financial-mathematics papers, from 1985, 2001, 1985 and 2013, compared with this kind of decomposition.





# Next Section – Clear Definitions

What is a scientific communication?

Clear Definitions

Example on 4 microstructure papers





# A proof is a route, not a destination

- The theorem is the **destination**; the proof is the **route**
- Two routes to the same place differ in *what you see and what you need* (Pythagoras: similar triangles vs. dissection)
- A journal judges the route: is it worth taking, which legs are fragile, where else can they be used?

**To speak about routes**

we need a formalism where a proof is an **object with parts**. Logic has one: **Curry–Howard**, where proofs are programs, lemmas are function calls, and chaining lemmas is composing functions.





# Curry–Howard for non-specialists

| Logic | Programming | In a paper |
|---|---|---|
| statement | type | a result |
| proof | program | an argument |
| using a lemma (cut) | calling a function | "by Lemma 2" |
| cut elimination | running ($\beta$-reduction) | **inlining every lemma** |
| unfolding a definition | $\delta$-reduction | **replacing names by definitions** |

↑ This is what an AI system does

Running always ends in the same direct proof (Church–Rosser, strong normalisation):
a **canonical** form, but one in which **the segments are gone**.





# Setting and Derivation-Level Intension

## Setting

- a formal language $L$, named axioms $\Sigma$, a proof calculus $K$
- a derivation: a graph of statements; leaves are axioms or hypotheses
- a paper has chosen results $T_1, \ldots, T_k$; keep only what leads to them: $D^{\mathrm{trim}}$
- definitions stay *folded*

## Same story: the relation $\approx$

$\approx$ is the smallest equivalence relation on derivations such that

1. isomorphic graphs are equivalent (reordering independent steps)
2. $D \approx D^{\mathrm{trim}}$ (dropping dead branches)
3. cut with an identity, re-association of cuts and packaging of conjunctions change nothing

and **nothing else**: no $\beta$-reduction, no $\delta$-unfolding

**Definition**

$$\mathrm{DLI}(D) = [D]_{\approx}, \qquad L_D = \bigcup_{S \in D^{\mathrm{trim}}} \mathrm{voc}(S)$$

the proof graph up to $\approx$, plus the vocabulary it uses

# Entering the formalisation: from a proof to its segments

1. A proof of $T$ from a hypothesis $S_0$ is an **arrow** $p : S_0 \to T$
   (derivations taken up to $\approx$; composing arrows = using a lemma)
2. Pick **intermediate results** $S_1, \ldots, S_{n-1}$ the author wants to expose and *cut* the arrow there:

$$S_0 \xrightarrow{p_1} S_1 \xrightarrow{p_2} \cdots \xrightarrow{p_n} S_n = T, \qquad p = p_n \circ \cdots \circ p_1$$

   This factorisation is a splitting $\sigma$.
3. Each $p_i : S_{i-1} \to S_i$ is a segment: a stand-alone lemma with its own hypothesis and conclusion.

**Pythagoras** $\top \xrightarrow{p_1} S_{\text{alt}} \wedge S_{\text{sim}} \xrightarrow{p_2} (a^2 = cd) \wedge (b^2 = ce) \xrightarrow{p_3} T_N$, so $n = 3$

**Why this works only without execution**

Running the proof ($\beta$-reduction) would merge all $p_i$ into one arrow $\top \to T_N$: the intermediate $S_i$ and the segments would be gone.





# Footprint: what a segment assumes

- Footprint $\mathrm{Ax}(D) \subseteq \Sigma$: the axioms (the background theory) at the leaves of the trimmed derivation, *as written*
  $\Rightarrow$ for a segment, $\mathrm{Ax}(p_i)$: what a reader who reuses $p_i$ must accept
- Footprints **add up** along a chain: $\mathrm{Ax}(q \circ p) = \mathrm{Ax}(p) \cup \mathrm{Ax}(q)$
  (running the proof can only give $\mathrm{Ax}(\mathrm{Norm}(q \circ p)) \subseteq \mathrm{Ax}(p) \cup \mathrm{Ax}(q)$, and hides who used what)
- Minimal footprints $\mathrm{Min}(S)$: the $\subseteq$-minimal axiom sets $\Sigma'$ with $\Sigma' \vdash S$ (not unique)

**Plain reading**

If some other proof $D'$ has $\mathrm{Ax}(D') \subsetneq \mathrm{Ax}(D)$, the proof **can be improved**: a **weak link**.
If $\mathrm{Ax}(D) \in \mathrm{Min}(S)$, nothing is assumed in excess: a candidate **strong link**.



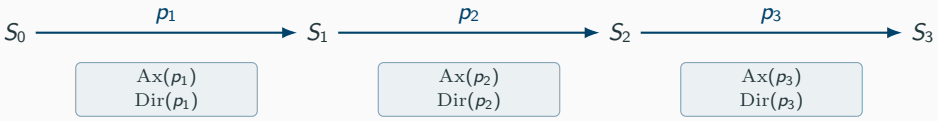


# The splitting $\sigma$: how the DE rebuilds everything

**Definition**

A splitting $\sigma$ of $p : S_0 \to T$ is a factorisation $p = p_n \circ \cdots \circ p_1$ at cut points $S_1, \ldots, S_{n-1}$ (below), each $S_i$ a statement of the trimmed derivation of $p$, each $p_i$ really using $S_{i-1}$.

in words: the milestones the author exposes, together with the segments between them




**What $(p, \sigma)$ lets us rebuild**

- **the proof**: compose the segments in the order of $\sigma$, $p = p_n \circ \cdots \circ p_1$ (up to $\approx$)
- **its assumptions**: $\mathrm{Ax}(p) = \bigcup_i \mathrm{Ax}(p_i)$
- **its reuse**: the meaning of $\mathrm{Dir}(p_i)$ is explained on the next slide.

Other splittings of the same $p$ rebuild the same proof but give other DEs: $\sigma$ records the author's choice.

# Directional Extension: segments, footprints, transport

$\mathrm{Ax}(p_i) \subseteq \Sigma$ is **the set of named axioms** occurring as leaves of the trimmed derivation of the segment $p_i$ (the hypothesis $S_{i-1}$ it starts from is not counted).

**Transport class** $\mathrm{Dir}(p_i)$: the ways to *carry* the segment $p_i$ into another proof or paper $(L', \Sigma')$.

It receives $p_i$ through an interpretation: a dictionary translating the symbols of $p_i$ into $L'$ such that every named leaf of $\mathrm{Ax}(p_i)$ stays valid in $\Sigma'$.

$\Rightarrow$ it *translates* the segment $p_i$ in $(L', \Sigma')$: $\mathrm{Dir}(p_i)$ collects its assumptions, and the dictionaries under which they hold elsewhere.

**Definition**

$$\mathrm{DE}(p, \sigma) = \big(\sigma,\ (\mathrm{Ax}(p_i))_{i \le n},\ (\mathrm{Dir}(p_i))_{i \le n}\big)$$

the cut points, what each segment assumes, where each segment can travel

*The same proof admits several splittings*, hence several DEs: $(p_1,\ p_2,\ p_3)$ has footprints: $(\{\alpha\}, \{\beta\}, \emptyset)$, $(p_1,\ p_3 \circ p_2)$ has footprints: $(\{\alpha\}, \{\beta\})$, $(p_2 \circ p_1,\ p_3)$ has footprints: $(\{\alpha, \beta\}, \emptyset)$.

The DLI fixes the available cut points; the DE records **which ones the author exposes**.





# Conclusion: Directional Extension = the route map

**Recall: Derived-Level Intension**

the route up to harmless changes

$$\mathrm{DLI}(D) = [D]_{\approx}, \quad L_D$$

it fixes **which milestones exist**

**Directional Extension**

$$\mathrm{DE}(p, \sigma) = \big(\sigma,\ (\mathrm{Ax}(p_i))_{i \le n},\ (\mathrm{Dir}(p_i))_{i \le n}\big)$$

- the milestones $\sigma$
- the luggage of each leg
- the places each leg can reach

**Summary**

The DLI fixes which milestones exist; the DE chooses which ones the author *publishes*, so that weak and strong legs can be seen by others.

A Lean script declares neither DLI nor DE.





# Back to our question: what does each AI role act on?

| | Literature search | Computations, solving steps | Exploring chains | Critical judgment | Full production |
|---|---|---|---|---|---|
| **Context** | ● | ○ | | ● | ○ |
| **Method** | | ○ | ● | ● | ● |
| **Arguments** | | ● | | | ● |
| **Watch out** | visibility bias | none for the story (same DLI, same DE) | weakest proof unnoticed (visible in DE) | uniformisation (extract DLI and DE) | a dream: intension always comes from humans |

● the AI acts on this part ○ this part stays with humans or is preserved blank: not concerned

*Context* = why this problem, which angle. *Method* = the chain of reasoning with weak and strong links. *Arguments* = the proofs of each step.

None of the roles removes the need to **declare the context (DLI) and the chain (DE)**.

## When DE is free, i.e. when proofs are abondant

- Before: the fixed cost of building a DE guaranteed some thinking behind the DLI
- Now: assert a theorem and a DLI, ask an AI for *some* true DE
- Likely outcome: only weak segments, thus not powerful DE, and nobody knows

**Consequences**

- Editors: demand an explicit DLI $\rightarrow$ judge the match with the scope of the journal
- Reviewers: demand DE with weak and strong link *to* scheck that weak links cannot easily be improved

Moreover:

**Training of young mathematicians**: practice stating intension (DLI) and structure (DE).





## Conclusion of this part

- AI is a tool: it changes the cost of proofs, not the role of mathematicians to **explore the map of possibles** such a way it support sciences (and mathematics) to create value.
- Abundant proofs judged only true/false are a *cloud of dust*
- Two objects to declare:
  - Derivation-Level Intension: problem and angle
  - Directional Extension: chain of reasoning with weak and strong links
- Scarce resource: a community's shared understanding of why a result matters and how to push it further

Lean versions of the propositions: `github.com/ca-lehalle/dlide`





# Next Section – Example on 4 microstructure papers

What is a scientific communication?

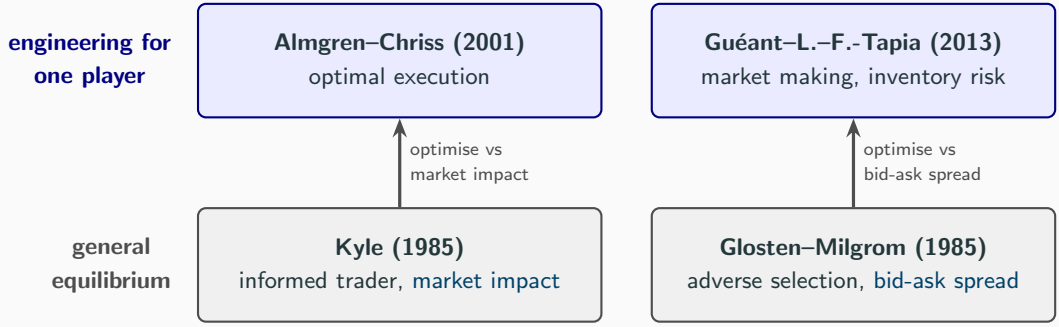
Clear Definitions

Example on 4 microstructure papers





# Test case: four microstructure papers

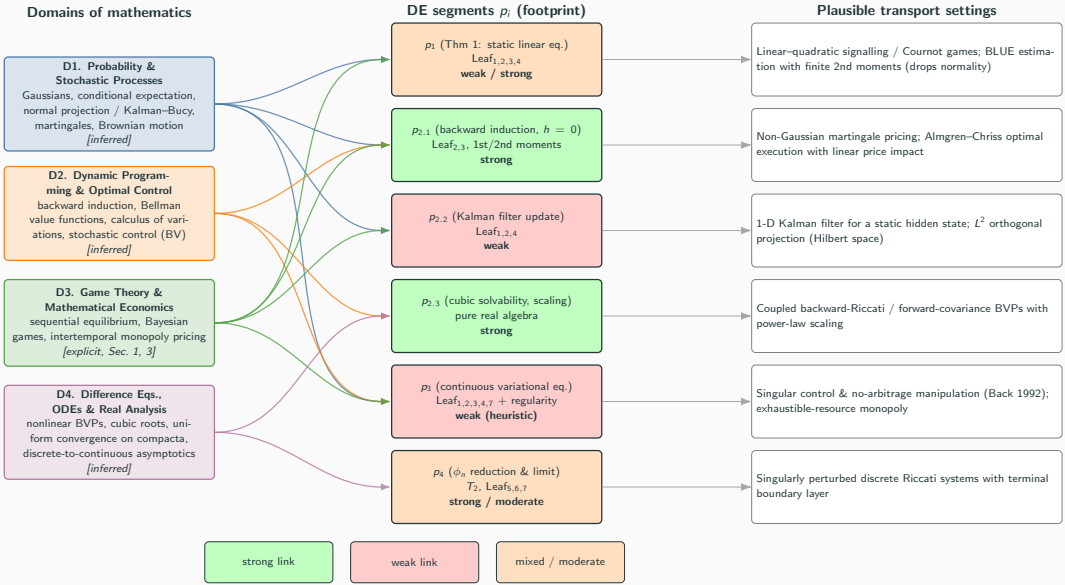



In each column: engineering optimisation sits on top of a price-formation model.

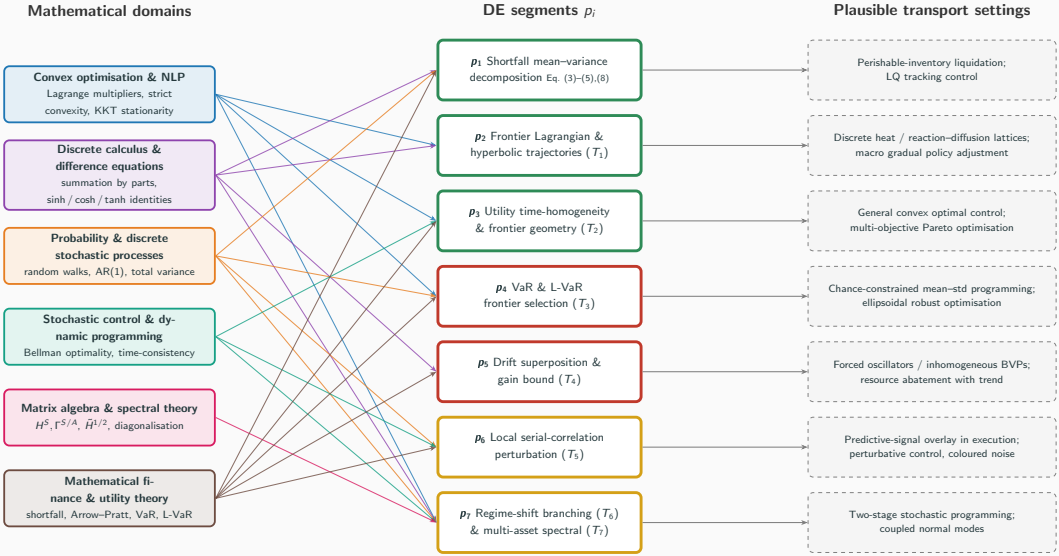
Next 4 slides:

- domains of mathematics $\rightarrow$ DE segments $p_i$ $\rightarrow$ transport settings
- Segment colour = link strength (strong / weak); same DLI/DE vocabulary as before

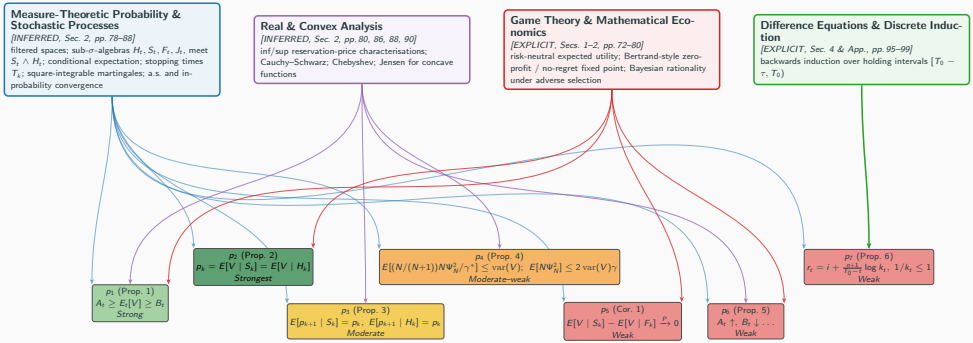
# Kyle (1985) – domains of mathematics used by the DE



# Almgren & Chriss (2001) – domains of mathematics used by the DE

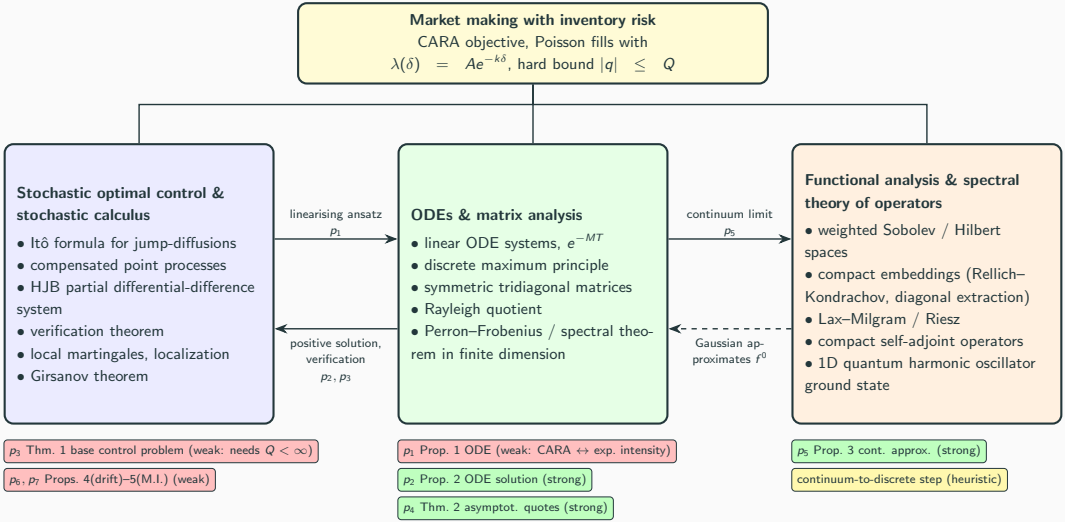


# Glosten & Milgrom (1985) – domains of mathematics used by the DE




**Edge key:** blue = probability (used by every segment: tower property, conditioning); purple = real/convex analysis (truncation lemma $p_1, p_4$; Cauchy–Schwarz $p_3$; Chebyshev $p_4$; Jensen $p_6$); red = game-theoretic equilibrium ($p_1, p_2$ fixed point; $p_3$ median normalisation; $p_6$ insider/uninformed mix); green = backwards induction ($p_7$).
**Segment colour** = DE link strength (dark green strongest → red weak).

# Guéant–L.–Fernandez-Tapia (2013) – domains of mathematics used by the DE



# From Kyle to Almgren–Chriss: what changes in the domains

**Dropped**

- **Game theory**: no more sequential equilibrium between informed trader and market makers
- **Gaussian filtering** (Kalman, normal projection)

**Kept, transformed**

- **Dynamic programming** and **difference equations**: backward induction becomes a deterministic convex program with sinh / cosh trajectories
- **Probability** shrinks to means and variances of a random walk

**New**

- **Convex optimisation** (Lagrangian, KKT)
- **Matrix and spectral** tools (multi-asset)
- **Utility / risk** (mean–variance, VaR)

**Reading**

The price impact, *derived* by equilibrium in Kyle, becomes a *parameter*: the footprint loses game theory and Gaussianity, and Kyle's strong segments (backward induction, algebra) are exactly those transported (his transport list already names optimal execution). The new weak links are on the **risk-objective side**: VaR selection and the drift bound.

# From Glosten–Milgrom to GLF: what changes in the domains

**Dropped**

- **Game theory**: zero-profit Bayesian specialist
- **Filtrations as information**: sub-$\sigma$-algebras, signals
- Jensen / Chebyshev style **real analysis**

**Kept, transformed**

- **Probability**: from information structure to **jump processes** in a fixed filtration
- **Induction**: discrete backward steps become an **HJB system**

**New**

- **Stochastic control** (Itô, verification, Girsanov)
- **Linear ODEs, tridiagonal matrices**
- **Functional analysis**: Sobolev spaces, compactness, spectral theory

**Reading**

Adverse selection is compressed into a *fill-intensity law* $\lambda(\delta) = Ae^{-k\delta}$: the spread becomes a control. Strong segments are the linear ODE reduction and the continuum limit; the weak ones sit at the **interface with the economics** (CARA $\leftrightarrow$ exponential intensity) and in the bound $|q| \leq Q$.

## Usefulness of the DE: lessons for market microstructure research

1. **Compare papers by footprint**: both engineering papers drop game theory and import optimisation, control or analysis; the economics survives as *one parameter or one law* (impact coefficient, $\lambda(\delta)$).

2. **Weak links show where to work**: they sit where the economic model meets the optimisation (risk objective, CARA/intensity link, bounded inventory). Relaxing them, e.g. *endogenising* impact or intensity, is the natural research direction.

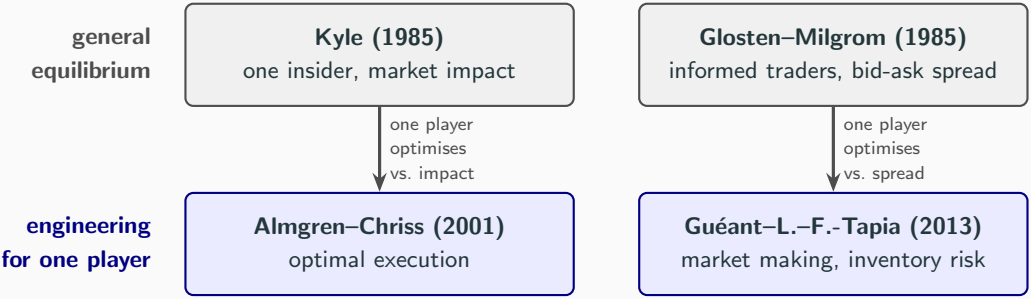
3. **Strong links show what to reuse**: backward induction, linear ODE / Riccati reductions and spectral arguments move between execution, market making and other settings.

4. **Footprints say what a generalisation costs**: e.g. Kyle's result needs only second moments, so Gaussianity can be dropped; GLF's $Q < \infty$ is a weak step.





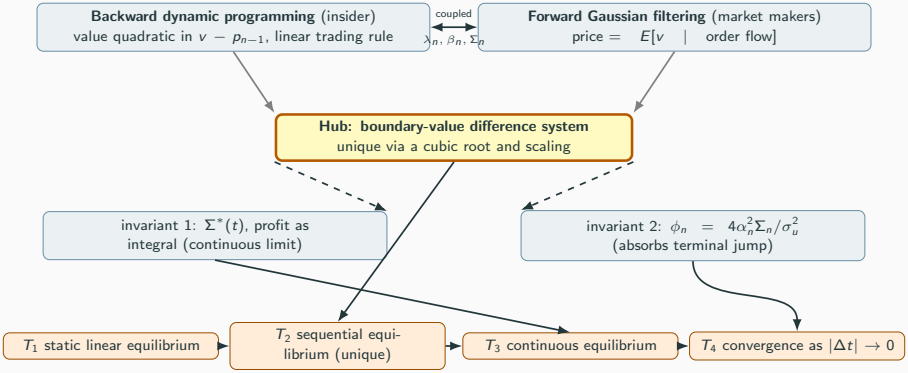
# DLI analysis of four microstructure papers




- The **DLI** of a paper is its *angle*: one sentence, a folded vocabulary, a hub step from which the results fan out
- Next 4 slides: one DLI each (Kyle, AC, GM, GLF); then what they share and what separates them

# Kyle (1985): two coupled engines

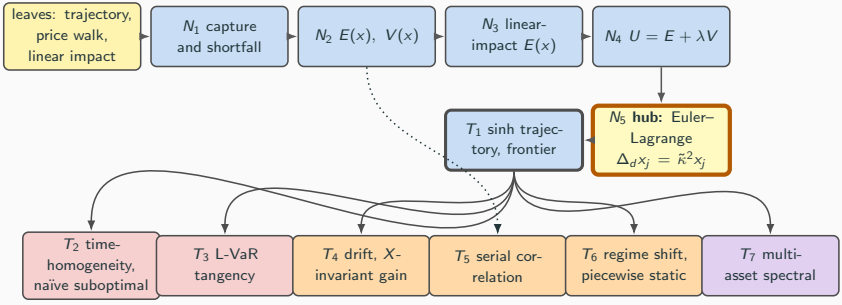
**Angle:** linear–quadratic dynamic programming coupled with Gaussian (Kalman) filtering; the insider's backward value function and the market makers' forward beliefs reduce to a single variable $\phi_n$.




Vocabulary: exogenous primitives and time grid · strategy and market variables · equilibrium concepts · structural parameters

# Almgren–Chriss (2001): one hub, a fan of results

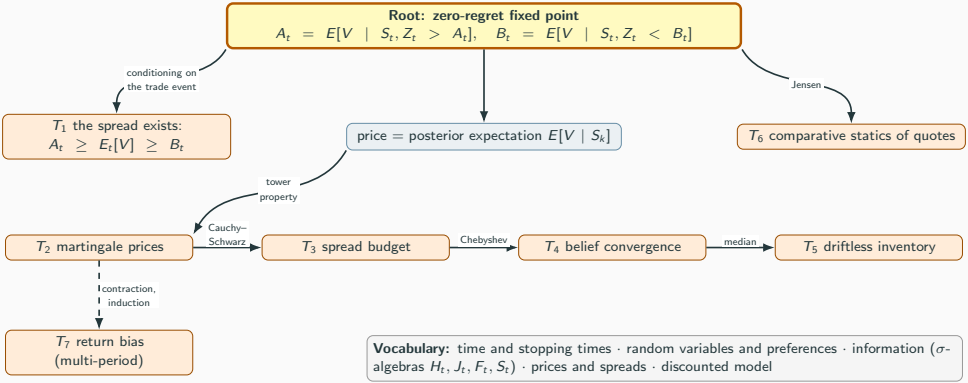
**Angle:** Markowitz-style mean–variance convex optimisation and discrete variational calculus: a static quadratic problem whose sinh trajectories form an efficient frontier; dynamics are closed-form perturbations.




**Folded vocabulary:** time, trajectory | price, impact | performance, frontier | risk, utility | information, perturbation | multi-asset

# Glosten–Milgrom (1985): one root mechanism

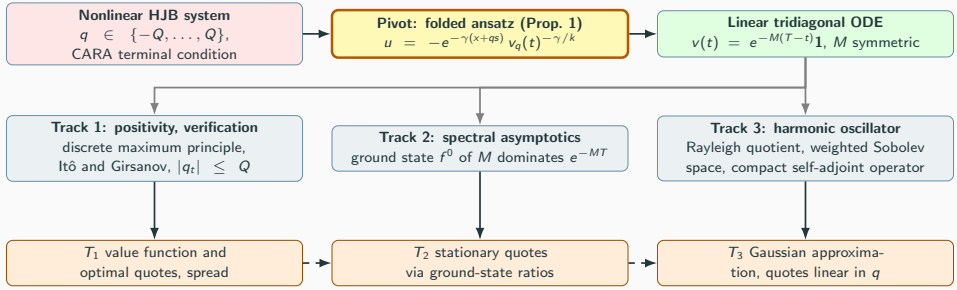
**Angle:** Bayesian martingale theory applied to sequential game-theoretic dealership pricing: quotes are conditional expectations that anticipate the information carried by the trade itself, not static markups.




**Vocabulary:** time and stopping times · random variables and preferences · information ($\sigma$-algebras $H_t, J_t, F_t, S_t$) · prices and spreads · discounted model

# Guéant–L.–Fernandez-Tapia (2013): one pivot, three tracks

**Angle:** exact linearisation and spectral reduction: a Cole–Hopf-type transformation turns the non-linear HJB problem on a bounded inventory lattice into a linear tridiagonal ODE system, whose long-time behaviour is a Perron–Frobenius ground state, approximated by a harmonic oscillator.




**Extensions** $T_4$, $T_5$: the same pipeline with a drift $\mu$ (extra term $-\beta q v_q$) or with market impact $\xi$ (ansatz with $+\frac{1}{2}\xi q^2$)

# What the four DLI share, family by family

**In all four:** a one-sentence angle, a folded vocabulary, and one hub from which the results fan out (Kyle: the boundary-value system; GM: the fixed point; AC: Euler–Lagrange; GLF: the folded ansatz).

**Driven by game theory** Kyle, GM

- the root is an **equilibrium condition**: profit maximisation + market efficiency (Kyle), zero-regret fixed point (GM)
- beliefs are **updated by conditioning**: Gaussian filter (Kyle), conditional expectation (GM)
- vocabulary: information and equilibrium concepts
- results: *properties of the equilibrium* (existence, uniqueness, martingale prices, convergence)

**Driven by optimisation** AC, GLF

- the root is an **objective**: mean–variance $U = E + \lambda V$ (AC), CARA value function (GLF)
- a **pivot makes it tractable**: Euler–Lagrange hub (AC), folded ansatz (GLF)
- tractable form: linear structure (sinh trajectories, linear ODE system)
- results: *optimal policies* (trajectory, quotes) and extensions from the same pipeline



# What separates the four DLI

## Between the two families

- **Who acts**: several players whose strategies must be consistent (Kyle, GM) vs. one player facing a given market (AC, GLF)
- **Market object**: derived in equilibrium (impact $\lambda$, spread $A_t - B_t$) vs. taken as a parameter or a law (linear impact, fill intensity)
- **Question**: "what is consistent?" vs. "what should I do?"

## Within each family

- **Kyle vs. GM**: Gaussian linear–quadratic dynamic programming and filtering, with a continuous limit; vs. conditional expectations on general information structures, a martingale spine, no Gaussian specialisation
- **AC vs. GLF**: static quadratic problem with closed-form perturbations; vs. stochastic control on a bounded inventory, linearised and then analysed spectrally





## Conclusion: reading papers by their angle

- Same market objects (impact, spread), **two angles**: equilibrium (Kyle, GM) and optimisation for one player (AC, GLF)
- The DLI summarises a paper by its angle, hub and vocabulary, so papers can be **compared before their proofs**
- The engineering papers compress the economics into one parameter or one law: the angle changed, not the market

**Next: the Directional Extension**

The DLI shows the shape of each story. The DE zooms into each hub: which segments are weak or strong, and where they can be reused.





# Backward Looking

What is a scientific communication?

Clear Definitions

- Derivation-Level Intension
- Directional Extension

Example on 4 microstructure papers

- Domains to Directional Extension tells the path from equilibrium to local optima
- Derived-Level Intension shows how papers communicate





# Overall Conclusion (1/2): From the Tree to the Forest

- **AI is not a menace:** Just as symbolic calculus freed us from routine hand computation, AI-assisted proving unlocks the bottleneck of long, human-driven technical derivations.
- **Technicalities must not be the tree that hides the forest:**
  - The **forest** is solving important problems that help scientists and mathematicians add value to our understanding of the world.
  - We do this by opening and comparing paths in the *Land of Possibles* [Bouveresse 1988].
- **A formal grammar that resists automation:**
  - **Derivation-Level Intension (DLI)** : encodes the chosen problem, perspective, and folded vocabulary.
  - **Directional Extension (DE)** : encodes the milestones, axiomatic footprints (**weak links** to improve), and transport classes (**strong links** to reuse).





# Overall Conclusion (2/2): Instrumenting DLI and DE

### For Journal Editors & Referees

- Move beyond proof-checking.
- Instrument **DLI** to assess scope and relevance, and **DE** to evaluate whether weak links can be strengthened and strong links transported.

### For Training Young Mathematicians

- Shift the focus earlier (before the PhD) from "producing a valid proof" to **identifying a DLI** and **designing different possible DEs**.

### Lifting Our Level of Abstraction

- Tested on four classics of **market microstructure** (Kyle, Glosten–Milgrom, Almgren–Chriss, GLF).
- DLI/DE decomposition gives a *bird's-eye view*: showing exactly how general equilibrium assumptions become engineering laws, and where the next research bottlenecks lie.

**Final word**

AI automates the *act of reasoning*; declaring **DLI** and **DE** preserves the *act of understanding*.

## Thank you for your attention...



**Any question?**

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# Reflexive appendix: the paper under its own DLI/DE

The formalism is applied to **this paper**: its trimmed derivation has **8 leaves** (setting and definitions), **2 well-definedness lemmas** and **4 shared results**; the DE cuts it into **6 segments** $p_1, \dots, p_6$.

- $\mathrm{L}_1$ calculus $K$: cut elimination ($\beta$), Church–Rosser, strong normalisation
- $\mathrm{L}_2$ DAG derivations, trimming, folded definitions (no $\delta$), $\mathrm{voc}(S)$
- $\mathrm{L}_3$ Def. 1: equivalence $\approx$
- $\mathrm{L}_4$ Def. 2: $\mathrm{DLI}(D) = [D]_{\approx}$, $L_D = \bigcup_{S \in D^{\mathrm{trim}}} \mathrm{voc}(S)$
- $\mathrm{L}_5$ $\mathrm{Norm}(D)$: class of the cut-free, $\delta$-folded form
- $\mathrm{L}_6$ Def. 3: proof category $\mathbf{P}(\Sigma)$, composition $=$ cut
- $\mathrm{L}_7$ Def. 4: footprint $\mathrm{Ax}(D) \subseteq \Sigma$
- $\mathrm{L}_8$ Def. 5: splitting $\sigma$ and $\mathrm{DE}(p, \sigma)$

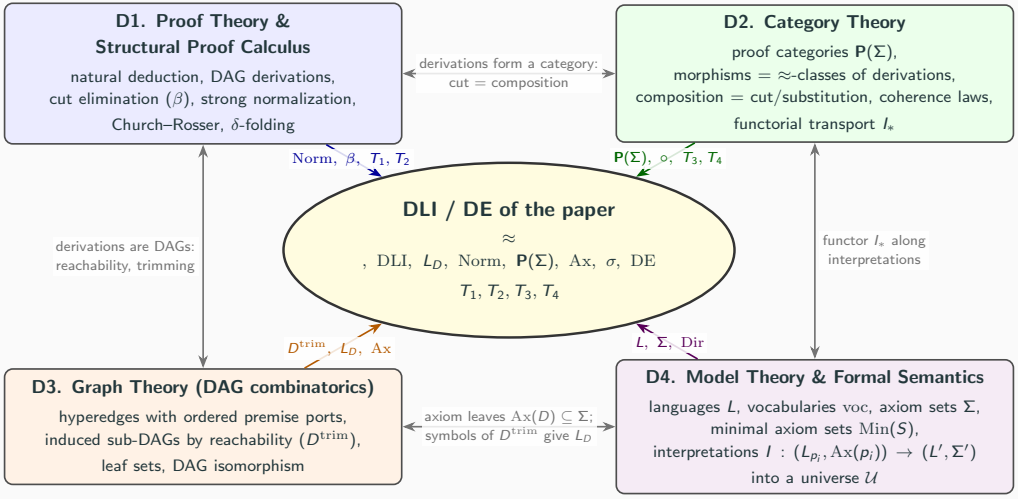
- $N_1$ $L_D$ is well defined on $\approx$-classes
- $N_2$ $\mathrm{Ax}(p)$ is well defined on morphisms of $\mathbf{P}(\Sigma)$
- $T_1$ Prop. 1(i): $\mathrm{DLI}(D_1) = \mathrm{DLI}(D_2) \Rightarrow \mathrm{Norm}(D_1) = \mathrm{Norm}(D_2)$
- $T_2$ Prop. 1(ii): same $\mathrm{Norm}$, different $\mathrm{DLI}$ (cut on a lemma whose vocabulary vanishes)
- $T_3$ Prop. 2(i): $\mathrm{Ax}(q \circ p) = \mathrm{Ax}(p) \cup \mathrm{Ax}(q)$; after $\mathrm{Norm}$ only $\subseteq$
- $T_4$ Prop. 2(ii): one $p$, several splittings, different footprint tuples

Next: the domains of mathematics, the DLI, the trimmed DAG, and the six segments.



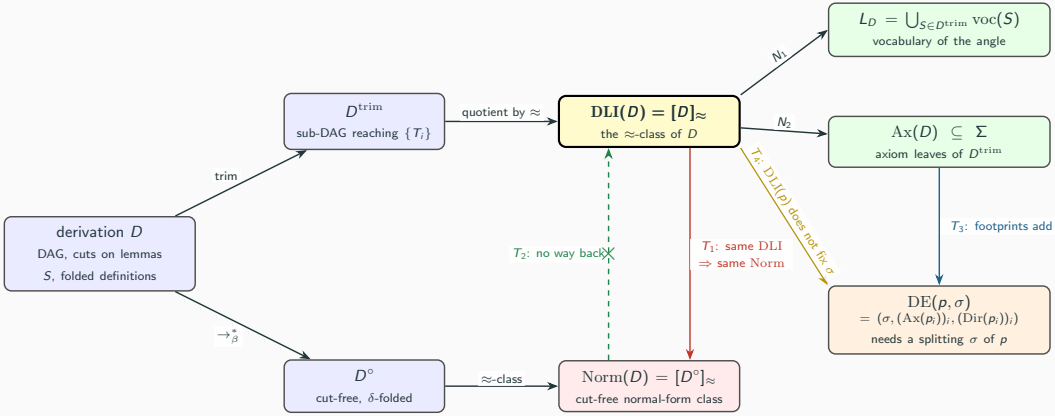


# Domains of mathematics required by the DE




All four domains are *explicit* in the formal core of the paper.

# The Derivation-Level Intension of the paper



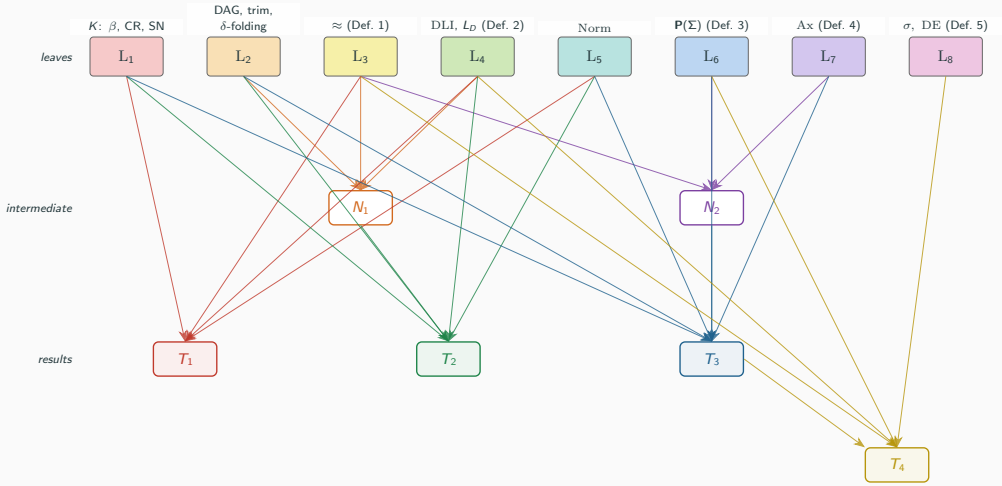

$\approx$: DAG isomorphism, trimming, identity cut, cut re-association, conjunction packaging; no general $\beta$, no $\delta$.

**Vocabulary loss ($T_2$):** if the cut lemma $S$ vanishes from $D^\circ$, then $L_D \neq L_{D^\circ}$, so $\mathrm{DLI}(D) \neq \mathrm{DLI}(D^\circ)$ although $\mathrm{Norm}(D) = \mathrm{Norm}(D^\circ)$.

C.-A. Lehalle

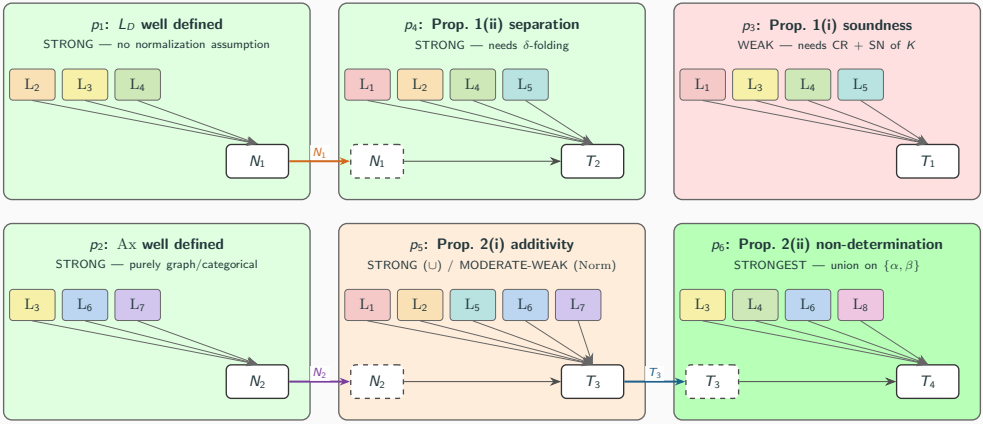


# The trimmed derivation DAG of the paper




Edge colour = the conclusion it feeds. Examples (Pythagoras) and the minimal-footprint remark are dropped by trimming.

# The DE of the paper: theorems as blocks, with their links




strong mixed weak strongest dashed box: imported result; thick arrows: links between blocks

Restrictive hypotheses sit in $p_3$ (Church–Rosser, strong normalisation) and $p_4$ ($\delta$-folding); the other segments only use DAG reachability, unions and composition.